

1.8 Introduction to Linear Transformations

I

A transformation (or function or mapping) T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each vector $\vec{x}$ in $\mathbb{R}^n$ a vector $T(\vec{x})$ in $\mathbb{R}^m$. The set $\mathbb{R}^n$ is called the domain of T , and $\mathbb{R}^m$ is called the codomain of T . The notation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ indicates that the domain of T is $\mathbb{R}^n$ and the codomain is $\mathbb{R}^m$. For $\vec{x}$ in $\mathbb{R}^n$, the vector $T(\vec{x})$ in $\mathbb{R}^m$ is called the image of $\vec{x}$ (under the action of T). The set of all images $T(\vec{x})$ is called the range of T .

Let's try to drive this home with an example.

$$\text{Let } A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}, \quad \vec{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}, \quad \vec{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}, \text{ and}$$

$(m \times n) \quad m \times n \quad m \times n \quad m \times n$
 $(3 \times 2) \quad (2 \times 1) \quad (3 \times 1) \quad (3 \times 1)$

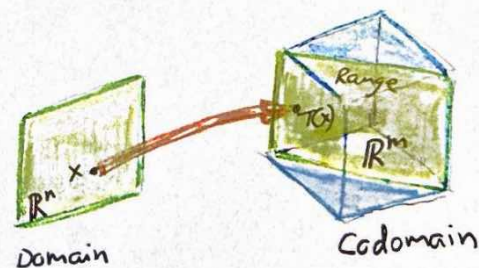
define a transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(\vec{x}) = A\vec{x}$, so that

$$T(\vec{x}) = A\vec{x} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -1x_1 + 7x_2 \end{bmatrix}$$

a. Find $T(\vec{u})$, the image of $\vec{u}$ under the transformation T .

To find $T(\vec{u})$ we compute $A\vec{u}$

$$T(\vec{u}) = A\vec{u} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 + 3 \\ 6 - 5 \\ -2 - 7 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$$



Domain, codomain and range of $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

For simplicity we denote matrix transformations as $\vec{x} \mapsto A\vec{x}$.

1.8 Introduction to Linear Transformations (cont'd)

II

b. Find an $\vec{x}$ in $\mathbb{R}^2$ whose image under T is $\vec{b}$.

They want us to solve $T(\vec{x}) = \vec{b}$ for $\vec{x}$. Since we have defined $T(\vec{x}) = A\vec{x}$, we will solve the equation

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$$

for $\vec{x}$ using elimination by matrix.
Setting up the augmented matrix and performing row reduction.

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{array} \right]$$

$$R_2 \leftarrow R_2 - 3R_1$$

$$R_3 \leftarrow R_3 + R_1$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & -2 \end{array} \right]$$

$$R_2 \leftarrow R_2 \cdot 1/14$$

$$R_3 \leftarrow R_3 \cdot 1/4$$

$$R_3 \leftarrow R_3 - R_2$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{array} \right]$$

$$R_1 \leftarrow R_1 + 3R_2$$

$$\left[\begin{array}{cc|c} 1 & 0 & 3/2 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{array} \right]$$

Row reduction gives

$$x_1 = 3/2$$

$$x_2 = -1/2$$

$$\boxed{\vec{x} = \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix}}$$

Testing result on A

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix} = \begin{bmatrix} 3/2 + 3/2 \\ 9/2 - 5/2 \\ -3/2 - 7/2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$$

This produces $\vec{b}$, which indicates the solution is correct.

We also take note of that we have no free variable, such that our solution

$$\vec{x} = \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix}$$

is the only solution to the equation

$$T(\vec{x}) = \vec{b}$$

1.8 Introduction to Linear Transformations (cont'd)

III

c. Is there more than one $\vec{x}$ whose image under T is $\vec{b}$? No, the solution we found in 5. is unique.

d. Determine if $\vec{c}$ is in the range of the transformation T .

What they're asking is if $A\vec{x} = \vec{c}$ is consistent, if it has a solution. If $\vec{c}$ is an image of $\vec{x}$ under the action of T . Let's set up an augmented matrix and check.

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{array} \right]$$

$$R_2 \leftarrow R_2 - 3R_1$$

$$R_3 \leftarrow R_3 + R_1$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{array} \right]$$

$$R_2 \leftarrow R_2 \cdot 1/14$$

$$R_3 \leftarrow R_3 \cdot 1/4$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 1 & 2 \end{array} \right]$$

$$R_3 \leftarrow R_3 - R_2$$

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 5/2 \end{array} \right]$$

Row reduction give us a last line that's impossible, $0 = 5/2$ is not valid.

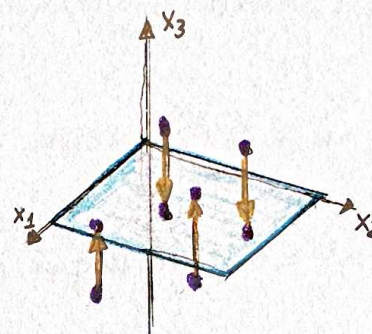
Thus, the system is inconsistent.

$\vec{c}$ is not in the range of T .

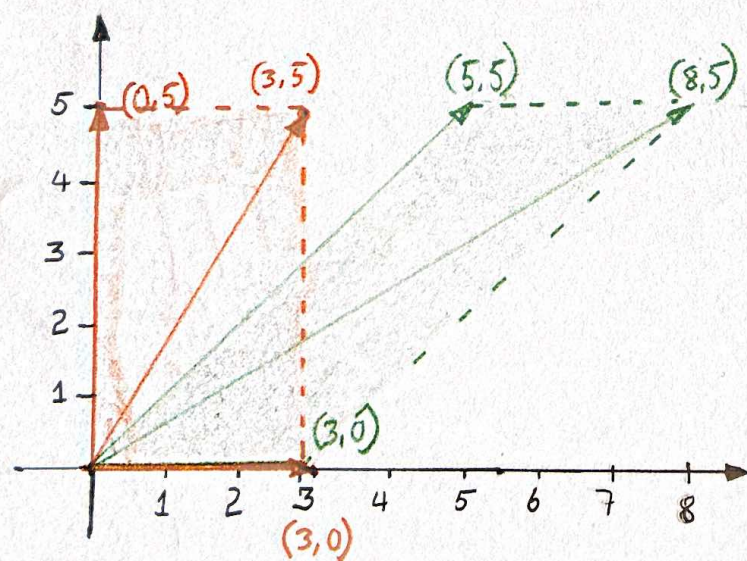
Example 2

If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, then the transformation $\vec{x} \mapsto A\vec{x}$ projects points in $\mathbb{R}^3$ onto the x_1x_2 -plane because

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix}$$



Example 3



Let $A = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$. The transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$T(\vec{x}) = A\vec{x}$ is called a shear transformation. It can be shown that if T acts on each point in the 3×5 rectangle to the left, then the set of images forms the sheared parallelogram.

Set $k=1$:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

The key idea is that the base of the rectangle stays, and every other point is sheared to one side.

The constant k , is called the shear factor.